

Problem Statement

The rapid expansion of cross-border banking networks (like Ecobank and Standard Bank), regional payment gateways, and pan-African e-commerce platforms has left support systems struggling to keep up. As digital trade accelerates across key hubs like Nigeria, Kenya, South Africa, and Ghana, customer support receives a high-volume, chaotic stream of unstructured messages daily. These complaints span cross-border merchant checkouts, peer-to-peer transfers, and multi-currency marketplace purchases. Today, institutions rely heavily on "overwhelmed" human customer service support because off-the-shelf English LLMs and rigid keyword chatbots consistently fail to parse localized financial dialects, code-switching (e.g., Pidgin and Sheng), and country-specific payment and e-commerce checkout rails (including NIP, M-Pesa, PayShap, and local card schemes).

We aim to develop a unified, cross-border complaint routing model optimized to ingest multi-channel text queues, parse regional code-switching, and accurately route issues to specialized financial and merchant back-office teams. The model will be fine-tuned on 13,000+ anonymized real-world African financial and digital commerce complaints sourced from Google Play Store applications across Nigeria, Kenya, South Africa, and Ghana. Leveraging the pre-trained multilingual foundation model Debarta-v3, the corpus will be classified into 10 operational taxonomies.

Successfully deploying this model will accelerate dispute resolution, eliminate backlogs, reduce cart abandonment, and protect capital for both consumers and cross-border merchants within Africa's high-stakes digital trade ecosystem.